Lab#9: Procedure and interrupts

Assignment #1: Write an assembler codes to accept from KBD X integer value and display on the screen the factorial of X, the program is to call procedure to find factorial X.

Solution:

```
ORG 100H
        MOV AH,09H
        LEA DX,MESSAGE
        INT 21H
        MOV AH,01H
        INT 21H
        MOV INPUT,AL
        SUB AL,"0"
        MOV BL,AL
        CALL FACTORIAL
        CALL ASCII_TO_INTEGER
        MOV AH,09H
        LEA DX,OUTPUT
        INT 21H
        CMP INPUT,"8"
        JAE END_PROGRAM
        MOV AH,09H
        LEA DX,Z_INTEGER
        INT 21H
END_PROGRAM:
HLT
    ASCII_TO_INTEGER PROC
        MOV BX,10
        MOV AX,RESULT
        MOV SI,3
C_LOOP: XOR DX,DX
        DIV BX
        ADD DL, '0'
        MOV Z_INTEGER[SI],DL
        DEC SI
        TEST AX,AX
        JNZ C_LOOP
    RET
    ASCII_TO_INTEGER ENDP
    FACTORIAL PROC
        MOV AX,1
    FAC:CMP BL,1
        JBE END_FAC
        MUL BX
        DEC BL
        JMP FAC
END_FAC:MOV RESULT,AX
        RET
    FACTORIAL ENDP
    MESSAGE DB "Enter The Number You Need To Find His Factorial : ",0AH,0DH,"$"
    INPUT DB ?
    OUTPUT DB 0AH,0DH,"The Result Is : ",0AH,0DH,"$"
    Z_INTEGER DB 5 DUP(?),"$"
    RESULT DW ?
```

ليث، عمرو

Assignment #2: Write an assembler codes to accept X string from KBD and copy characters in X that have ones in BIT#5 to Y string and display it on the screen

```
ORG 100H

        MOV AH,09H
        LEA DX,MESSAGE
        INT 21H

        MOV AH,0AH
        LEA DX,INPUT
        INT 21H

        MOV SI,2
        MOV DI,0
        MOV CX,10
L1:     TEST INPUT[SI],20H
        JNZ ONES
        INC SI
        LOOP L1
        JMP END_L
ONES:
        MOV AL,INPUT[SI]
        MOV Y[DI],AL
        INC DI
        INC SI
        LOOP L1
END_L:
        MOV AH,09H
        LEA DX,SPACE
        INT 21H


        MOV AH,09H
        LEA DX,Y
        INT 21H

HLT

    MESSAGE DB "INPUT STRING [MAXIMUM 10 CHARS] ",0AH,0DH,"$"
    INPUT DB 10,?,10 DUP("$")
    SPACE DB 0AH,0AH,0DH,"$"
    Y DB 10 DUP("$")
```

لیث، عمرو